Algoma UNIVERSITY

School of Computer Science & Technology - Algoma

Linear Algebra

Math 1057

Instructor

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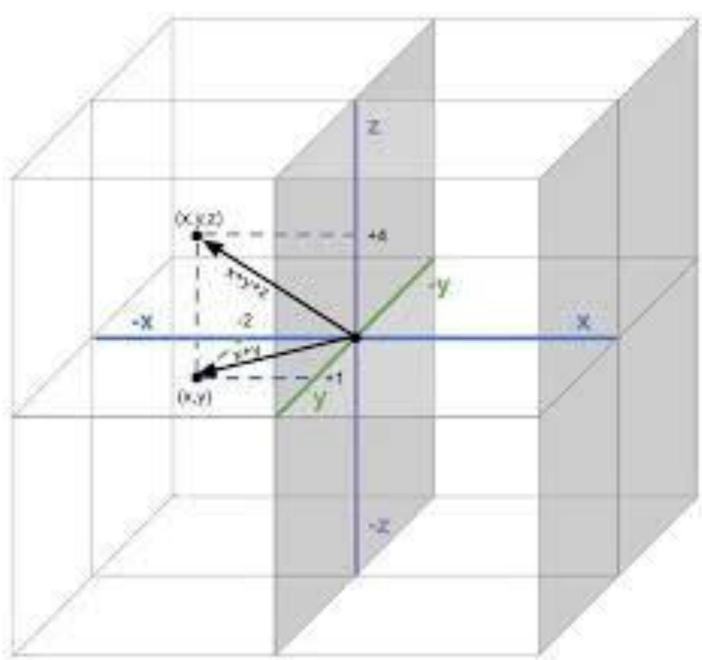

20 Inner Products and Orthogonality

Lecture 20

Inner Products and Orthogonality

20.1 Inner Product on $\mathbb{R}^n$

The inner product on $\mathbb{R}^n$ generalizes the notion of the **dot product** of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ that you may are already familiar with.

Definition 20.1: Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and let $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{R}^n$. The **inner product** of $\mathbf{u}$ and $\mathbf{v}$ is

$$\mathbf{u} \bullet \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

Notice that the inner product $\mathbf{u} \bullet \mathbf{v}$ can be computed as a matrix multiplication as follows:

$$\mathbf{u} \bullet \mathbf{v} = \mathbf{u}^T \mathbf{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

The following theorem summarizes the basic algebraic properties of the inner product.

Theorem 20.2: Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^n$ and let α be a scalar. Then

(a) $\mathbf{u} \bullet \mathbf{v} = \mathbf{v} \bullet \mathbf{u}$

(b) $(\mathbf{u} + \mathbf{v}) \bullet \mathbf{w} = \mathbf{u} \bullet \mathbf{w} + \mathbf{v} \bullet \mathbf{w}$

(c) $(\alpha \mathbf{u}) \bullet \mathbf{v} = \alpha(\mathbf{u} \bullet \mathbf{v}) = \mathbf{u} \bullet (\alpha \mathbf{v})$

(d) $\mathbf{u} \bullet \mathbf{u} \geq 0$, and $\mathbf{u} \bullet \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

Example 20.3. Let $\mathbf{u} = (2, -5, -1)$ and let $\mathbf{v} = (3, 2, -3)$. Compute $\mathbf{u} \bullet \mathbf{v}$, $\mathbf{v} \bullet \mathbf{u}$, $\mathbf{u} \bullet \mathbf{u}$, and $\mathbf{v} \bullet \mathbf{v}$.

Solution. By definition:

$$\begin{aligned}
\mathbf{u} \bullet \mathbf{v} &= (2)(3) + (-5)(2) + (1)(-3) = -1 \\
\mathbf{v} \bullet \mathbf{u} &= (3)(2) + (2)(-5) + (-3)(1) = -1 \\
\mathbf{u} \bullet \mathbf{u} &= (2)(2) + (-5)(-5) + (-1)(-1) = 30 \\
\mathbf{v} \bullet \mathbf{v} &= (3)(3) + (2)(2) + (-3)(-3) = 22.
\end{aligned}$$

□

We now define the length or norm of a vector in $\mathbb{R}^n$.

Definition 20.4: The **length** or **norm** of a vector $\mathbf{u} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \bullet \mathbf{u}} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}.$$

A vector $\mathbf{u} \in \mathbb{R}^n$ with norm 1 will be called a **unit vector**:

$$\|\mathbf{u}\| = 1.$$

Below is an important property of the inner product.

Theorem 20.5: Let $\mathbf{u} \in \mathbb{R}^n$ and let α be a scalar. Then

$$\|\alpha\mathbf{u}\| = |\alpha|\|\mathbf{u}\|.$$

Proof. We have

$$\begin{aligned}
\|\alpha\mathbf{u}\| &= \sqrt{(\alpha\mathbf{u}) \bullet (\alpha\mathbf{u})} \\
&= \sqrt{\alpha^2(\mathbf{u} \bullet \mathbf{u})} \\
&= |\alpha|\sqrt{\mathbf{u} \bullet \mathbf{u}} \\
&= |\alpha|\|\mathbf{u}\|.
\end{aligned}$$

□

By Theorem 20.5, any non-zero vector $\mathbf{u} \in \mathbb{R}^n$ can be scaled to obtain a new unit vector in the same direction as $\mathbf{u}$. Indeed, suppose that $\mathbf{u}$ is non-zero so that $\|\mathbf{u}\| \neq 0$. Define the new vector

$$\mathbf{v} = \frac{1}{\|\mathbf{u}\|}\mathbf{u}$$

Notice that $\alpha = \frac{1}{\|\mathbf{u}\|}$ is just a scalar and thus $\mathbf{v}$ is a scalar multiple of $\mathbf{u}$. Then by Theorem 20.5 we have that

$$\|\mathbf{v}\| = \|\alpha\mathbf{u}\| = |\alpha| \cdot \|\mathbf{u}\| = \frac{1}{\|\mathbf{u}\|} \cdot \|\mathbf{u}\| = 1$$

and therefore $\mathbf{v}$ is a unit vector, see Figure 20.1. The process of taking a non-zero vector $\mathbf{u}$ and creating the new vector $\mathbf{v} = \frac{1}{\|\mathbf{u}\|}\mathbf{u}$ is sometimes called **normalization** of $\mathbf{u}$.

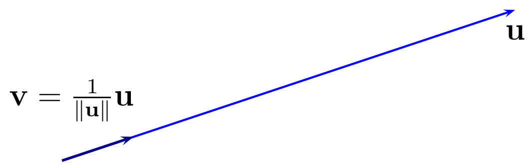

Figure 20.1: Normalizing a non-zero vector.

Example 20.6. Let $\mathbf{u} = (2, 3, 6)$. Compute $\|\mathbf{u}\|$ and find the unit vector $\mathbf{v}$ in the same direction as $\mathbf{u}$.

Solution. By definition,

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \bullet \mathbf{u}} = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7.$$

Then the unit vector that is in the same direction as $\mathbf{u}$ is

$$\mathbf{v} = \frac{1}{\|\mathbf{u}\|}\mathbf{u} = \frac{1}{7}\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 2/7 \\ 3/7 \\ 6/7 \end{bmatrix}$$

Verify that $\|\mathbf{v}\| = 1$:

$$\|\mathbf{v}\| = \sqrt{(2/7)^2 + (3/7)^2 + (6/7)^2} = \sqrt{4/49 + 9/49 + 36/49} = \sqrt{49/49} = \sqrt{1} = 1.$$

□

Now that we have the definition of the length of a vector, we can define the notion of distance between two vectors.

Definition 20.7: Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in $\mathbb{R}^n$. The **distance between u and v** is the length of the vector $\mathbf{u} - \mathbf{v}$. We will denote the distance between $\mathbf{u}$ and $\mathbf{v}$ by $\mathrm{d}(\mathbf{u}, \mathbf{v})$. In other words,

$$\mathrm{d}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

Example 20.8. Find the distance between $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 7 \\ -9 \end{bmatrix}$.

Solution. We compute:

$$\mathrm{d}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(3-7)^2 + (-2+9)^2} = \sqrt{65}.$$

□

20.2 Orthogonality

In the context of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$, orthogonality is synonymous with perpendicularity. Below is the general definition.

Definition 20.9: Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are said to be **orthogonal** if $\mathbf{u} \bullet \mathbf{v} = 0$.

In $\mathbb{R}^2$ and $\mathbb{R}^3$, the notion of orthogonality should be familiar to you. In fact, using the Law of Cosines in $\mathbb{R}^2$ or $\mathbb{R}^3$, one can prove that

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\| \cdot \|\mathbf{v}\| \cos(\theta) \tag{20.1}$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$. If $\theta = \frac{\pi}{2}$ then clearly $\mathbf{u} \bullet \mathbf{v} = 0$. In higher dimensions, i.e., $n \geq 4$, we can use equation (20.1) to *define* the angle between vectors $\mathbf{u}$ and $\mathbf{v}$. In other words, the **angle** between any two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is define to be

$$\theta = \arccos\left(\frac{\mathbf{u} \bullet \mathbf{v}}{\|\mathbf{u}\| \cdot \|\mathbf{v}\|}\right).$$

The general notion of orthogonality in $\mathbb{R}^n$ leads to the following theorem from grade school.

Theorem 20.10: (Pythagorean Theorem) Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Solution. First recall that $\|\mathbf{u} + \mathbf{v}\| = \sqrt{(\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v})}$ and therefore

$$\begin{aligned}
\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v}) \\
&= \mathbf{u} \bullet \mathbf{u} + \mathbf{u} \bullet \mathbf{v} + \mathbf{v} \bullet \mathbf{u} + \mathbf{v} \bullet \mathbf{v} \\
&= \|\mathbf{u}\|^2 + 2(\mathbf{u} \bullet \mathbf{v}) + \|\mathbf{v}\|^2.
\end{aligned}$$

Therefore, $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ if and only if $\mathbf{u} \bullet \mathbf{v} = 0$. □

We now introduce orthogonal sets.

Definition 20.11: A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is said to be an **orthogonal set** if any pair of distinct vectors $\mathbf{u}_i, \mathbf{u}_j$ are orthogonal, that is, $\mathbf{u}_i \bullet \mathbf{u}_j = 0$ whenever $i \neq j$.

In the following theorem we prove that orthogonal sets are linearly independent.

Theorem 20.12: Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ be an orthogonal set of non-zero vectors in $\mathbb{R}^n$. Then the set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is linearly independent. In particular, if $p = n$ then the set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is basis for $\mathbb{R}^n$.

Solution. Suppose that there are scalars $c_1, c_2, \ldots, c_p$ such that

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p = \mathbf{0}.$$

Take the inner product of $\mathbf{u}_1$ with both sides of the above equation:

$$c_1(\mathbf{u}_1 \bullet \mathbf{u}_1) + c_2(\mathbf{u}_2 \bullet \mathbf{u}_1) + \cdots + c_p(\mathbf{u}_p \bullet \mathbf{u}_1) = \mathbf{0} \bullet \mathbf{u}_1.$$

Since the set is orthogonal, the left-hand side of the last equation simplifies to $c_1(\mathbf{u}_1 \bullet \mathbf{u}_1)$. The right-hand side simplifies to $\mathbf{0}$. Hence,

$$c_1(\mathbf{u}_1 \bullet \mathbf{u}_1) = \mathbf{0}.$$

But $\mathbf{u}_1 \bullet \mathbf{u}_1 = \|\mathbf{u}_1\|^2$ is not zero and therefore the only way that $c_1(\mathbf{u}_1 \bullet \mathbf{u}_2) = \mathbf{0}$ is if $c_1 = 0$. Repeat the above steps using $\mathbf{u}_2, \mathbf{u}_3, \ldots, \mathbf{u}_p$ and conclude that $c_2 = 0$, $c_3 = 0, \ldots, c_p = 0$. Therefore, $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is linearly independent. If $p = n$, then the set $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is automatically a basis for $\mathbb{R}^n$. $\square$

Example 20.13. Is the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ an orthogonal set?

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \ \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \ \mathbf{u}_3 = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$$

Solution. Compute

$$\begin{aligned}
\mathbf{u}_1 \bullet \mathbf{u}_2 &= (1)(0) + (-2)(1) + (1)(2) = 0 \\
\mathbf{u}_1 \bullet \mathbf{u}_3 &= (1)(-5) + (-2)(-2) + (1)(1) = 0 \\
\mathbf{u}_2 \bullet \mathbf{u}_3 &= (0)(-5) + (1)(-2) + (2)(1) = 0
\end{aligned}$$

Therefore, $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal set. By Theorem 20.12, the set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is linearly independent. To verify linear independence, we computed that $\det(\begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3\end{bmatrix}) = 30$, which is non-zero. $\square$

We now introduce orthonormal sets.

Definition 20.14: A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is said to be an **orthonormal set** if it is an orthogonal set and if each vector $\mathbf{u}_i$ in the set is a unit vector.

Consider the previous orthogonal set in $\mathbb{R}^3$:

$$\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}.$$

It is not an orthonormal set because none of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are unit vectors. Explicitly, $\|\mathbf{u}_1\| = \sqrt{6}$, $\|\mathbf{u}_2\| = \sqrt{5}$, and $\|\mathbf{u}_3\| = \sqrt{30}$. However, from an **orthogonal set** we can create an **orthonormal set** by normalizing each vector. Hence, the set

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \begin{bmatrix} 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}, \begin{bmatrix} -5/\sqrt{30} \\ -2/\sqrt{30} \\ 1/\sqrt{30} \end{bmatrix} \right\}$$

is an orthonormal set.

20.3 Coordinates in an Orthonormal Basis

As we will see in this section, a basis $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ of $\mathbb{R}^n$ that is also an orthonormal set is highly desirable when performing computations with coordinates. To see why, let $\mathbf{x}$ be any vector in $\mathbb{R}^n$ and suppose we want to find the coordinates of $\mathbf{x}$ in the basis $\mathcal{B}$, that is we seek to find $[\mathbf{x}]_{\mathcal{B}} = (c_1, c_2, \ldots, c_n)$. By definition, the coordinates $c_1, c_2, \ldots, c_n$ satisfy the equation

$$\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_n\mathbf{u}_n.$$

Taking the inner product of $\mathbf{u}_1$ with both sides of the above equation and using the fact that $\mathbf{u}_1 \bullet \mathbf{u}_2 = 0$, $\mathbf{u}_1 \bullet \mathbf{u}_3 = 0$, and $\mathbf{u}_1 \bullet \mathbf{u}_n = 0$, we obtain

$$\mathbf{u}_1 \bullet \mathbf{x} = c_1(\mathbf{u}_1 \bullet \mathbf{u}_1) = c_1(1) = c_1$$

where we also used the fact that $\mathbf{u}_i$ is a unit vector. Thus, $c_1 = \mathbf{u}_1 \bullet \mathbf{x}$! Repeating this procedure with $\mathbf{u}_2, \mathbf{u}_3, \ldots, \mathbf{u}_n$ we obtain the remaining coefficients $c_2, \ldots, c_n$:

$$\begin{aligned} c_2 &= \mathbf{u}_2 \bullet \mathbf{x} \\ c_3 &= \mathbf{u}_3 \bullet \mathbf{x} \\ \vdots &= \ \vdots \\ c_n &= \mathbf{u}_n \bullet \mathbf{x}. \end{aligned}$$

Our previous computation proves the following theorem.

Theorem 20.15: Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be an orthonormal basis for $\mathbb{R}^n$. The coordinate vector of $\mathbf{x}$ in the basis $\mathcal{B}$ is

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} \mathbf{u}_1 \bullet \mathbf{x} \\ \mathbf{u}_2 \bullet \mathbf{x} \\ \vdots \\ \mathbf{u}_n \bullet \mathbf{x} \end{bmatrix}.$$

Hence, computing coordinates with respect to an orthonormal basis can be done without performing any row operations and all we need to do is compute inner products! We make the important observation that an alternate expression for $[\mathbf{x}]_{\mathcal{B}}$ is

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} \mathbf{u}_1 \bullet \mathbf{x} \\ \mathbf{u}_2 \bullet \mathbf{x} \\ \vdots \\ \mathbf{u}_n \bullet \mathbf{x} \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \mathbf{x} = \mathbf{U}^T \mathbf{x}$$

where $\mathbf{U} = [\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \cdots \;\; \mathbf{u}_n]$. On the other hand, recall that by definition $[\mathbf{x}]_{\mathcal{B}}$ satisfies $\mathbf{U}[\mathbf{x}]_{\mathcal{B}} = \mathbf{x}$, and therefore $[\mathbf{x}]_{\mathcal{B}} = \mathbf{U}^{-1}\mathbf{x}$. If we compare the two identities

$$[\mathbf{x}]_{\mathcal{B}} = \mathbf{U}^{-1}\mathbf{x} \quad \text{and} \quad [\mathbf{x}]_{\mathcal{B}} = \mathbf{U}^T\mathbf{x}$$

we suspect then that $\mathbf{U}^{-1} = \mathbf{U}^T$. This is indeed the case. To see this, let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ be an orthonormal basis for $\mathbb{R}^n$ and put

$$\mathbf{U} = [\mathbf{u}_1 \; \mathbf{u}_2 \; \cdots \; \mathbf{u}_n].$$

Consider the matrix product $\mathbf{U}^T\mathbf{U}$, and recalling that $\mathbf{u}_i \bullet \mathbf{u}_j = \mathbf{u}_i^T \mathbf{u}_j$, we obtain

$$\begin{aligned} \mathbf{U}^T\mathbf{U} &= \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{u}_1^T\mathbf{u}_1 & \mathbf{u}_1^T\mathbf{u}_2 & \cdots & \mathbf{u}_1^T\mathbf{u}_n \\ \mathbf{u}_2^T\mathbf{u}_1 & \mathbf{u}_2^T\mathbf{u}_2 & \cdots & \mathbf{u}_2^T\mathbf{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_n^T\mathbf{u}_1 & \mathbf{u}_n^T\mathbf{u}_2 & \cdots & \mathbf{u}_n^T\mathbf{u}_n \end{bmatrix} \\ &= \mathbf{I}_n. \end{aligned}$$

Therefore,

$$\mathbf{U}^{-1} = \mathbf{U}^T.$$

A matrix $\mathbf{U} \in \mathbb{R}^{n\times n}$ such that

$$\mathbf{U}^T\mathbf{U} = \mathbf{U}\mathbf{U}^T = \mathbf{I}_n$$

is called a **orthogonal matrix**. Hence, if $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\}$ is an orthonormal set then the matrix

$$\mathbf{U} = \begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n\end{bmatrix}$$

is an orthogonal matrix.

Example 20.16. Consider the vectors

$$\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\end{bmatrix},\ \mathbf{v}_2 = \begin{bmatrix}-1\\4\\1\end{bmatrix},\ \mathbf{v}_3 = \begin{bmatrix}2\\1\\-2\end{bmatrix},\ \mathbf{x} = \begin{bmatrix}1\\2\\-1\end{bmatrix}.$$

(a) Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal basis for $\mathbb{R}^3$.
(b) Then, if necessary, normalize the basis vectors $\mathbf{v}_i$ to obtain an orthonormal basis $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ for $\mathbb{R}^3$.
(c) For the given $\mathbf{x}$ find $[\mathbf{x}]_{\mathcal{B}}$.

Solution. (a) We compute that $\mathbf{v}_1 \bullet \mathbf{v}_2 = 0$, $\mathbf{v}_1 \bullet \mathbf{v}_3 = 0$, and $\mathbf{v}_2 \bullet \mathbf{v}_3 = 0$, and thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal set. Since orthogonal sets are linearly independent and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ consists of three vectors then $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is basis for $\mathbb{R}^3$.
(b) We compute that $\|\mathbf{v}_1\| = \sqrt{2}$, $\|\mathbf{v}_2\| = \sqrt{18}$, and $\|\mathbf{v}_3\| = 3$. Then let

$$\mathbf{u}_1 = \begin{bmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\end{bmatrix},\ \mathbf{u}_2 = \begin{bmatrix}-1/\sqrt{18}\\4/\sqrt{18}\\1/\sqrt{18}\end{bmatrix},\ \mathbf{u}_3 = \begin{bmatrix}2/3\\1/3\\-2/3\end{bmatrix}$$

Then $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is now an orthonormal set and thus since $\mathcal{B}$ consists of three vectors then $\mathcal{B}$ is an orthonormal basis of $\mathbb{R}^3$.
(c) Finally, computing coordinates in an orthonormal basis is easy:

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix}\mathbf{u}_1 \bullet \mathbf{x}\\\mathbf{u}_2 \bullet \mathbf{x}\\\mathbf{u}_3 \bullet \mathbf{x}\end{bmatrix} = \begin{bmatrix}0\\2/\sqrt{18}\\5/3\end{bmatrix}$$

□

Example 20.17. The standard unit basis

$$\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\} = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\ \begin{bmatrix}0\\1\\0\end{bmatrix},\ \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$$

in $\mathbb{R}^3$ is an orthonormal basis. Given any $\mathbf{x} = (x_1, x_2, x_3)$, we have $[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}$. On the other hand, clearly

$$\begin{aligned} x_1 &= \mathbf{x} \bullet \mathbf{e}_1 \\ x_2 &= \mathbf{x} \bullet \mathbf{e}_2 \\ x_3 &= \mathbf{x} \bullet \mathbf{e}_3 \end{aligned}$$

Example 20.18. (Orthogonal Complements) Let W be a subspace of $\mathbb{R}^n$. The **orthogonal complement** of W, which we denote by $\mathsf{W}^\perp$, consists of the vectors in $\mathbb{R}^n$ that are orthogonal to every vector in W. Using set notation:

$$\mathsf{W}^\perp = \{\mathbf{u} \in \mathbb{R}^n \,:\, \mathbf{u} \bullet \mathbf{w} = 0 \text{ for every } \mathbf{w} \in \mathsf{W}\}.$$

(a) Show that $\mathsf{W}^\perp$ is a subspace.
(b) Let $\mathbf{w}_1 = (0, 1, 1, 0)$, let $\mathbf{w}_2 = (1, 0, -1, 0)$, and let $\mathsf{W} = \operatorname{span}\{\mathbf{w}_1, \mathbf{w}_2\}$. Find a basis for $\mathsf{W}^\perp$.

Solution. (a) The vector $\mathbf{0}$ is orthogonal to every vector in $\mathbb{R}^n$ and therefore it is certainly orthogonal to every vector in W. Thus, $\mathbf{0} \in \mathsf{W}^\perp$. Now suppose that $\mathbf{u}_1, \mathbf{u}_2$ are two vectors in $\mathsf{W}^\perp$. Then for any vector $\mathbf{w} \in \mathsf{W}$ it holds that

$$(\mathbf{u}_1 + \mathbf{u}_2) \bullet \mathbf{w} = \mathbf{u}_1 \bullet \mathbf{w} + \mathbf{u}_2 \bullet \mathbf{w} = 0 + 0 = 0.$$

Therefore, $\mathbf{u}_1 + \mathbf{u}_2$ is also orthogonal to $\mathbf{w}$ and since $\mathbf{w}$ is an arbitrary vector in W then $(\mathbf{u}_1 + \mathbf{u}_2) \in \mathsf{W}^\perp$. Lastly, let α be any scalar and let $\mathbf{u} \in \mathsf{W}^\perp$. Then for any vector $\mathbf{w}$ in W we have that

$$(\alpha\mathbf{u}) \bullet \mathbf{w} = \alpha(\mathbf{u} \bullet \mathbf{w}) = \alpha \cdot 0 = 0.$$

Therefore, $\alpha\mathbf{u}$ is orthogonal to $\mathbf{w}$ and since $\mathbf{w}$ is an arbitrary vector in W then $(\alpha\mathbf{u}) \in \mathsf{W}^\perp$. This proves that $\mathsf{W}^\perp$ is a subspace of $\mathbb{R}^n$.

(b) A vector $\mathbf{u} = (u_1, u_2, u_3, u_3)$ is in $\mathsf{W}^\perp$ if $\mathbf{u} \bullet \mathbf{w}_1 = 0$ and $\mathbf{u} \bullet \mathbf{w}_2 = 0$. In other words, if

$$\begin{aligned} u_2 + u_3 &= 0 \\ u_1 - u_3 &= 0 \end{aligned}$$

This is a linear system for the unknowns u_1, u_2, u_3, u_4. The general solution to the linear system is

$$\mathbf{u} = t \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

Therefore, a basis for $\mathsf{W}^\perp$ is $\{(1, 0, 1, 0), (0, 1, -1, 0)\}$.

□

After this lecture you should know the following:

- how to compute inner products, norms, and distances
- how to normalize vectors to unit length
- what orthogonality is and how to check for it
- what an orthogonal and orthonormal basis is
- the advantages of working with orthonormal basis when computing coordinate vectors